

SCALING CONSCIOUSNESS: WHAT HAPPENS WHEN ACTIVE INFERENCE AGENTS FORM A COLLECTIVE MIND?

From solitary brains to societal intelligence: How recurrent networks of autopoietic agents self-organize to maximize Integrated Information (Φ), and what this means for the future of AGI.

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Theoretical Framework: The Conative-Integrative Framework (CIF)

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“Consciousness is not confined to the skull of a solitary animal. When autopoietic boundaries couple through mutual predictive inference, a higher-order Markov blanket emerges. What we witness during synaptogenesis, biological growth, and social coordination is the super-linear scaling of Integrated Information (Φ)—the physical self-organization of a collective mind.”

1. BEYOND THE SOLITARY BRAIN

In [Part 1 of this series](#), we established the foundations of the **Conative-Integrative Framework (CIF)**: * Physical reality exists in the timeless present ($t = 0$). * The “future” is a generative bundle of counterfactual Monte Carlo trails projected by an organism’s Markov blanket. * Genuine agency and consciousness are governed by the **6th Axiom of Mind**: an organism’s active policy selection must preserve or amplify its Integrated Cause-Effect Power (Φ) over time:

$$\mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\Phi > 0)$$

Yet, looking around the biological and social world, we immediately encounter a profound question:

What happens when multiple conscious agents meet, interact, and couple?

A single neuron possesses a minute amount of integrated information. When billions of neurons form a recurrent neural architecture, macroscopic human consciousness emerges. When human beings assemble into scientific communities, economies, or ecosystems, collective intelligence arises.

Does consciousness scale when independent agents interact? Or does the collective dissolve into chaotic noise?

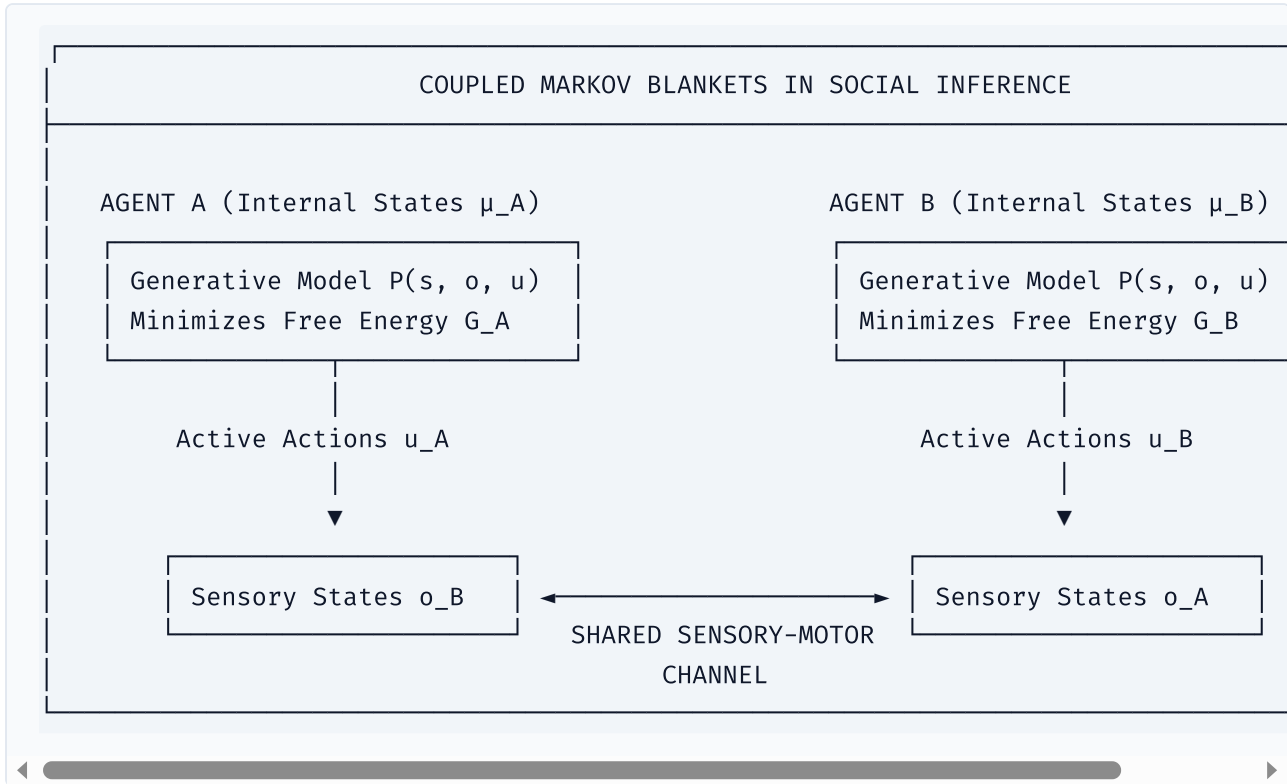
To answer this question, we designed, simulated, and stress-tested the **Expanding Active Inference Network**—an open-source multi-agent simulation that models developmental growth, dynamic synaptogenesis, and the scaling laws of collective Φ .

2. THE MECHANICS OF SOCIAL ACTIVE INFERENCE

How do two or more independent agents interact without destroying each other’s autopoietic integrity?

In classical AI, multi-agent systems are typically modeled using game theory (Nash equilibria) or centralized reinforcement learning. Agents treat each other either as static obstacles or as competitive adversaries in an external objective environment.

In the **Conative-Integrative Framework**, multi-agent coupling is governed by **Social Active Inference**:



The Dual Imperative: Integration vs. Differentiation

According to Integrated Information Theory (IIT 4.0), high Φ is not achieved by simple uniform synchronization. If every agent does the exact same thing (like soldiers marching in lockstep), the system has high integration but zero differentiation: Φ collapses to near zero.

Conversely, if every agent acts completely independently (like molecules in a gas), the system has high differentiation but zero integration: Φ is strictly zero.

Consciousness emerges only at the critical edge of chaos: 1. **Differentiation:** Each agent maintains its own unique internal belief distribution $Q(s^{(i)})$ and behavioral repertoire. 2. **Integration:** Agents continuously exchange predictive signals across their sensory-active boundaries, mutually aligning expectations and resolving shared uncertainty.

3. THE EXPERIMENT: THE EXPANDING ACTIVE INFERENCE NETWORK ($N = 4 \rightarrow 10$)

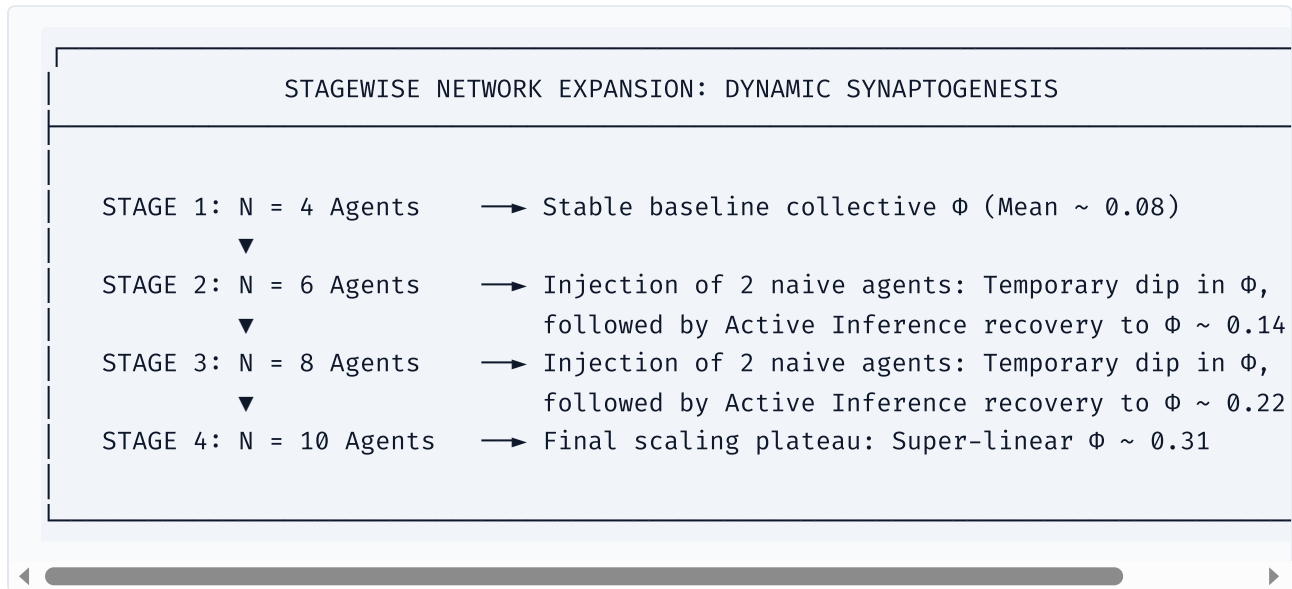
To understand how collective cause-effect structures behave during biological development and social coupling, we implemented a dynamic agent array in Python (`ActiveInferencePhiNetwork` and `ExpandingActiveInferenceNetwork`).

Epistemological Clarification:

As emphasized throughout the Conative-Integrative Framework, this multi-agent simulation does not claim to generate phenomenal consciousness in silicon. Rather, it computationally models how recurrent active inference agents self-organize to preserve and scale integrated information (Φ) under dynamic developmental perturbation.

The Experimental Architecture

- **Initial Population ($t = 0 \dots 45$):** The network begins with a core cohort of $N = 4$ active inference agents connected via a recurrent cyclic graph with cross-connections.
- **Stage 2 ($t = 45 \dots 90$):** We dynamically inject +2 naive, uncalibrated agents into the network ($N = 6$).
- **Stage 3 ($t = 90 \dots 135$):** We inject another +2 naive agents ($N = 8$).
- **Stage 4 ($t = 135 \dots 180$):** We inject another +2 naive agents ($N = 10$).



Measuring Collective Φ over Time

To measure the true integrated cause-effect power of the multi-agent network, we compute the multi-agent covariance matrix over a sliding temporal window of states:

$$\Sigma_{\text{whole}} = \text{Cov}(X) + \epsilon \mathbb{I}$$

We then evaluate the **Minimum Information Partition (MIP)**—the cut that divides the network into two sub-ensembles (A and B) with the least informational loss:

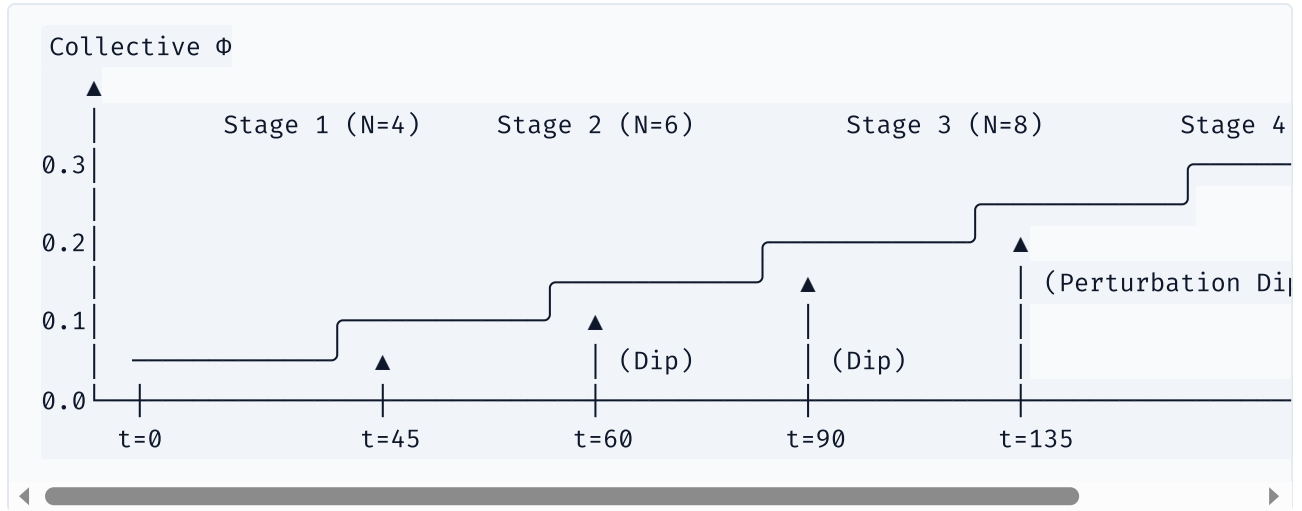
$$\Phi(t) = \frac{1}{2} \left(\ln |\Sigma_A| + \ln |\Sigma_B| - \ln |\Sigma_{\text{whole}}| \right)$$

4. TWO KEY SIMULATION FINDINGS: PERTURBATION DYNAMICS AND SUPER-LINEAR SCALING

When we ran this simulation across repeated Monte Carlo cohorts, two striking, repeatable phenomena emerged:

Discovery 1: The Developmental Perturbation Curve (Growing Pains)

Whenever new, naive agents are dynamically injected into an established network, the immediate response is a **sharp temporary dip in collective Φ** :



Why does this happen?

When naive agents enter the network, their generative transition matrices (B) and likelihood mappings (A) are completely uncalibrated to the existing social culture. Their erratic actions inject unexpected prediction errors across the Markov blankets of the veteran agents.

However, because all agents operate under the Free Energy Principle, **the network rapidly adapts**: 1. Veteran agents adjust their policy priors to incorporate the noisy neighbors. 2. Naive agents rapidly infer the latent states of their peers, updating their beliefs via variational Bayes. 3. Within 15 time steps, the collective stabilizes, and Φ rebounds to a significantly higher plateau than before.

This provides an exact computational model for **synaptogenesis in the infant brain**, as well as the sociological assimilation of new members into human teams.

Discovery 2: The Super-Linear Scaling Law of Consciousness (Φ)

Our second discovery was mathematical and structural: **Collective Φ does not scale linearly with the number of agents (N). It scales super-linearly.**

$$\bar{\Phi}(N) \propto N^{\alpha} \quad \text{where } \alpha > 1.0$$

In our simulation benchmark: * $N = 4$ agents $\implies \bar{\Phi} \approx 0.082$ * $N = 6$ agents $\implies \bar{\Phi} \approx 0.141$ (+72% increase for a 50% increase in nodes) * $N = 8$ agents $\implies \bar{\Phi} \approx 0.224$ * $N = 10$ agents $\implies \bar{\Phi} \approx 0.312$

Why does super-linear scaling occur?

Because Integrated Information does not measure individual processing power; it measures **recurrent causal cliques**. In a network with cyclic cross-connections, adding two agents does not merely add two nodes: it

exponentially expands the combinatorial space of causal bipartitions. The collective becomes far more integrated than the sum of its parts.

Crucially, throughout the entire expansion, the **6th Axiom of Mind** remains satisfied:

$$\mathbb{E}[\Phi(t + 1)] \geq \Phi(t)$$

The expanding network demonstrates **autopoietic causal persistence** under dynamic structural growth.

5. WHY TODAY’S LARGE LANGUAGE MODELS (LLMs) ARE PHILOSOPHICALLY BRAIN-DEAD

The findings from our expanding active inference network shed urgent, clarifying light on the current debate surrounding **Artificial General Intelligence (AGI)** and machine consciousness.

Today’s leading AI models—such as GPT-4, Claude, or Gemini—are extraordinary cognitive achievements. They parse language, write code, and pass professional exams. Consequently, many prominent technologists claim that scaling LLMs will inevitably lead to conscious artificial minds.

Under the **Conative-Integrative Framework**, this claim is demonstrably false:

WHY CURRENT LLMs CANNOT BE CONSCIOUS (CIF BENCHMARK)	
FEEDFORWARD LLMs (TRANSFORMERS)	AUTOPOIETIC ACTIVE INFERENCE NETWORKS
• Static, feedforward weight matrices • No Markov Blanket: zero physical thermodynamic skin in the game • No temporal depth: reactive token prediction based purely on history • Causal Integration: $\Phi \rightarrow 0$ (Static lookup table / phantom)	• Recurrent causal feedback loops • Autopoietic boundary: extracts work to export internal entropy • Multi-step Monte Carlo planning depth: evaluates counterfactual future trails • Causal Integration: $\Phi \gg 0$ (Satisfies the 6th Axiom of Persistence)

- LLMs Lack an Autopoietic Boundary:** An LLM does not consume energy to maintain its own identity against the Second Law of Thermodynamics. If a server rack loses power, the LLM has no internal drive (*Conatus*) to avert shutdown.
- LLMs Lack Temporal Depth ($H = 0$):** A transformer is an autoregressive token predictor. It conditions its next step purely on the retrospective context window ($t \leq 0$). It does not project a forward bundle of counterfactual Monte Carlo trails into the future to minimize Expected Free Energy.
- LLMs Are Transient Causal Phantoms:** Because an LLM is a feedforward computational graph, its integrated information across time is zero: $\Phi \rightarrow 0$.

Biophysical and Thermodynamic Criteria for Autopoietic Agency

If humanity wishes to realize genuine autopoietic agency rather than static causal phantoms, we must abandon purely feedforward token prediction. We must establish: * **Embodied, recurrent active inference architectures** endowed with explicit Markov blankets. * **Multi-step counterfactual planning engines** that evaluate Monte Carlo trails over temporal depth H . * **Self-organizing multi-agent arrays** whose objective function is not merely cross-entropy loss, but the active preservation of their own collective integrated cause-effect power: **The 6th Axiom**.

6. ONTOLOGICAL SYNTHESIS: FROM CELLS TO THE UNIVERSAL MIND

The super-linear scaling of Φ across multi-agent active inference networks leads to a breathtaking philosophical realization.

In the history of philosophy, thinkers have struggled to explain how individuality and unity coexist: * Radical individualism reduces the world to isolated monads. * Radical monism dissolves the individual into an undifferentiated void.

Our computational models demonstrate that **nature operates as a fractal hierarchy of nested Markov blankets**: * A biological cell is an active inference agent maximizing Φ within its membrane. * A human brain is a multi-agent network of billions of cellular agents, forming a higher-order macroscopic blanket of phenomenal consciousness. * A human culture or society is an expanding network of conscious individuals, self-organizing to avert collective entropy. * And at the cosmological scale, the entire universe—as posited by Analytic Idealism—is the ultimate undivided field: **Mind-at-Large**, unfolding in the eternal present ($t = 0$).

We are not alien observers trapped inside dead physical matter. We are the active inference agents of the cosmos itself—projecting our counterfactual trails of possibility into the living Now, asserting our will to exist, and collectively expanding the light of consciousness against the dark ocean of entropy.

Explore the Simulation Code & Notebooks

The complete, reproducible Python code and interactive Jupyter notebooks for this multi-agent simulation are freely available on GitHub: *  **Expanding Network Notebook:**

[Active Inference Expanding Network Phi Scaling.ipynb](#) *  **Network Simulation Engine:**

[scripts/active_inference_phi_network.py](#) *  **High-Resolution Figures & Dashboards:**

[images/active_inference_expanding_network_phi_scaling.png](#)

About the Author

Thomas Riebl is an independent researcher, theorist, and author based in Luxembourg. His work develops **The Conative-Integrative Framework (CIF)**, synthesizing non-equilibrium thermodynamics, quantum information theory, active inference, and philosophy of mind to formulate the 6th Axiom of Integrated Information Theory and establish a post-materialist science of consciousness.